Public Opinion Quarterly, Vol. 81, No. 1, Spring 2017, pp. 1–36

NONRESPONSE BIAS FOR UNIVARIATE AND MULTIVARIATE ESTIMATES OF SOCIAL ACTIVITIES AND ROLES

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STANLEY PRESSER

Abstract Nonresponse bias is a fundamental concern for survey researchers, as understanding when and to what extent it occurs is critical to producing accurate statistics. According to the social integration hypothesis, individuals who participate in a broad range of social activities and roles should be more likely to respond to surveys (Goyder 1987; Groves and Couper 1998). As a result, prevalence estimates of social activities and roles should be upwardly biased. By contrast, models predicting these activities and roles may be unbiased if the nonrespondents are missing at random, as the results of Abraham, Helms, and Presser 2009 suggest. Using the rich frame information available on the American Time Use Survey (ATUS) and the Survey of Health, Ageing, and Retirement in Europe (SHARE) Wave II, we compare the full sample to the respondent sample on 30 different social roles and activities. The results suggest that nonresponse bias was widespread and often large on univariate estimates, but was usually small in multivariate models and typically did not alter the inferences drawn from such models.

Introduction

Researchers have tried to explain nonresponse bias since the inception of probability-based surveys. Identifying the types of estimates that suffer from nonresponse bias may be useful in judging the quality of the estimates as well as in targeting efforts to correct the bias. Studies of nonresponse bias have often proposed that individuals who are more socially integrated (i.e., individuals

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doi:10.1093/poq/nfw037 Advance Access publication October 11, 2016

who participate in a broad range of social relationships) are more likely to respond to a survey request while individuals who are socially isolated are less likely to participate (Goyder 1987; Groves and Couper 1998).

There are three grounds for the hypothesis that integration is related to survey response. First, individuals wish to fit in with their social groups. If individuals perceive that survey participation is consistent with the expectations of group members or if they believe that other members would participate, they will be more likely to agree to be respondents (Groves, Cialdini, and Couper 1992; Brehm 1993; Groves and Couper 1998). Second, people may participate to avoid consequences (perceived or real) of not participating. Individuals who are socially integrated internalize a set of social norms learned from their group membership(s). Failure to comply with these norms may result in a sense of guilt, and individuals may weigh such guilt as a negative consequence and factor it into their response decision. Third, individuals who are socially integrated may be more likely to feel that their participation in a survey will yield longer-term personal or group benefits (e.g., safer neighborhoods or more school funding) (Groves, Cialdini, and Couper 1992; Goyder, Boyer, and Martinelli 2006). Individuals are more likely to perceive a social benefit or assign a greater weight to the benefit if they feel connected to the group(s) that will profit from the information gained from the survey.

Various research has shown support for the social integration hypothesis. For example, individuals who read the newspaper, practiced sports, watched television, were interested in politics, traveled, and/or volunteered were all found to be significantly more likely to agree to complete the Dutch Time Use Survey (Van Ingen, Stoop, and Breedveld 2009). Similarly, Groves, Singer, and Corning (2000) showed that individuals who were civically engaged were more likely to respond to a Detroit Area Study follow-up survey. Other researchers have used the social integration hypothesis to explain why social activities such as volunteering and attending church appear related to survey cooperation (Woodberry 1998; Groves, Singer, Corning 2000; Abraham, Helms, and Presser 2009). Finally, single-person households, renters, and individuals out of the labor force—all of which have been interpreted as proxies for weaker social integration—are less likely to participate in surveys than their counterparts (Abraham, Maitland, and Bianchi 2006).

Since integrated individuals have been found to be more likely to respond to a survey request and since integrated individuals are more likely to participate in any given social relationship (e.g., voting, playing a team sport, attending religious services), it follows that survey estimates of social activities and social roles should be upwardly biased. Consistent with this, Kennickell (2005) reported that survey estimates of charitable contributions were much higher than the corresponding population estimates. Similarly, Abraham, Helms, and Presser (2009) found that survey estimates of volunteering were inflated by nonresponse. By contrast, Smith (1984) generally found only small differences between initially cooperative respondents and those who initially refused on frequency of visiting with friends, visiting with neighbors, visiting with relatives, going to a bar, visiting with parents, and visiting siblings, with the direction of the bias not consistent across measures.

The research to date has several limitations. First, studies have often examined only a single social relationship, for example volunteering. Second, studies that have looked at a variety of measures have not had access to the values for both respondents and nonrespondents and instead have used late respondents or converted refusers as proxies for nonrespondents (e.g., Smith 1984). Finally, formal tests have rarely been used to test whether the differences between respondents and the full sample are statistically significant. Instead, researchers have tested for differences between respondents and nonrespondents, which demonstrates whether the two groups are different from each other but not whether the estimate among respondents is statistically different from the full sample.

In this paper, we used data from respondents and nonrespondents to investigate the extent of nonresponse bias on a wide variety of social activity and role indicators. We test three hypotheses. The first hypothesis is that measures of social activities and social roles should be upwardly biased as a result of differential nonresponse among individuals of various integration levels (H1).

The second hypothesis stems from the fact that there are many routes one may take to become integrated. For example, one individual may be politically engaged by boycotting, voting, or running for city council. Another individual may be connected in other ways—playing a team sport and hosting dinner parties. Individuals act in accordance with the expectations of the social groups in which they belong. The expectation of survey participation is greater among civically and politically oriented groups, since the request to participate in a survey is most similar to work done in these groups: volunteering of time, helping the greater good, and cooperating with a government (in the case of a government-sponsored survey) request. Thus, the second hypothesis (H2) is that variables measuring political and social activities and roles should suffer from higher levels of nonresponse bias than other activity and role measures.

The third hypothesis builds on the fact that bias in univariate estimates does not necessarily imply that multivariate estimates will be biased. Assume, for example, a model in which the likelihood of voting is a function of race. Given that isolated individuals are less likely to respond, then integrated individuals will be overrepresented in the survey. Since integrated individuals are more likely to vote, the survey would overestimate the proportion of voters, biasing the intercept. However, if socially isolated Whites were equally as likely as socially isolated non-Whites to be nonrespondents, then the effect of race on voting would be unbiased.

Indeed, Abraham, Helms, and Presser (2009) reported that volunteering was more common among respondents, across many different subgroups. Their analysis suggests that a researcher may estimate multivariate models without concern for nonresponse bias, though their work was based on a single measure (volunteering) and compared respondents to nonrespondents, not respondents to the full sample. Thus, our third hypothesis (H3) is that the coefficients of independent variables in multivariate models predicting social activities and roles will not suffer from nonresponse bias.

DATA

We used two datasets that have information about a diverse set of social activities and roles for both respondents and nonrespondents: the 2012 American Time Use Survey (ATUS) and Wave II of the Survey of Health, Ageing and Retirement in Europe (SHARE).

American Time Use Survey (ATUS)

The ATUS sample was drawn from former Current Population Survey (CPS) participating households. The CPS frame was constructed from the 2000 Decennial Census, an area frame, and an update of new housing permits. The CPS sample was a multistage stratified sample covering 824 individual sampling areas. Once a household was sampled, a roster was completed and a household member aged 15 or older was selected to complete the interview.[1]

Two months after "retirement" from CPS (i.e., aging out of the panel after eight data collection waves), ATUS staff randomly drew 2,190 CPS households for ATUS. This occurred monthly throughout 2012, resulting in an ATUS 2012 sample of 26,280 former CPS households. A three-stage design corrected for CPS's small-state oversample and introduced an oversample of racial/ethnic minorities and households with children. ATUS then randomly selected an individual 15 years or older from the household roster collected in the CPS.

A bilingual pre-notification letter[2] was mailed to all selected individuals, who were randomly assigned to a reference day, called on the following day, and asked about the reference day's activities. Callbacks were made as necessary over a period of eight weeks.[3] This resulted in an AAPOR Response Rate 2 of 53.2 percent on the 2012 ATUS (American Association for Public Opinion Research 2016; Bureau of the Census 2014).

In order to test our hypotheses, measures of social roles and social activities were needed for both respondents and nonrespondents. These were collected in the November 2011 CPS as part of the Civic Engagement (CE) Supplement administered at the end of the main CPS interview.

While the November CPS achieved a response rate of 90.6 percent, the CE Supplement captured only 81.6 percent of CPS respondents, resulting in an

1. For more information on CPS frame construction and sampling, please see chapter 3 of Technical Paper 66 (Bureau of the Census 2006).
2. Letters were mailed and interviews were conducted in English and Spanish. No other languages were used.
3. For the approximately 5 percent of households that did not have a telephone number available on the frame, the pre-notification letters were mailed with $40 inactive debit cards and instructions to call the telephone center. Upon call-in, the interviewer provided the PIN to activate the card.

overall response rate of 73.9 percent. While this is still higher than most surveys, research suggests that the CPS overestimates employment and healthcare coverage and undercounts several minorities (Schmitt and Baker 2006). If any of these variables are correlated with integration or if less-integrated individuals were less apt to participate in the CPS and/or the CE Supplement, there would be an upward skew in the prevalence estimates of the proportion of individuals who participate in various social roles and activities in the sample frame. The true population prevalence would be lower than the estimate calculated from the full sample, weakening the association between nonresponse and social integration.

All analyses were limited to 5,150 cases (2,779 ATUS respondents and 2,371 nonrespondents).[4] To be included, an individual had to have completed the CE Supplement (i.e., proxy respondents were excluded) and have been the sampled household member for ATUS.[5] The resulting sample was disproportionately female, renters, college educated, and non-Hispanic White. Included individuals were also less likely to be married, reported a lower household income, were older, and lived in smaller households. Despite these differences, all analyses between respondents and the analytic subsample were unbiased, since comparisons were made between the subsample and the subsample ATUS respondents.

A total of 18 indicators of social activities and social roles were available for both respondents and nonrespondents, 10 of which measured civic and political activities or roles (table 1). Eleven of the variables were dichotomous, while the remaining seven variables were ordinal. In analyses using proportions, categorical variables were collapsed into "never" and "at least once." When analyses were conducted using the ordinal categories, some categories had to be collapsed due to small cell sizes. The categories used in the analyses are shown in the last column in table 1.

The measure of "parent" did not include all parents, only adults who had children under 18 living in the household. As a result, a grandmother whose grandchildren live with her would have been labeled a parent, whereas a mother of an adult child would not. While this deviated from the traditional definition of "parent," it was appropriate in this context. Being a parent is a societal role and integrates an individual through parent–teacher interactions, parent–parent interactions, and strengthening bonds with extended family. This effect is generally strongest when the child is a minor and lives at home.

4. This suggests a response rate of 54.0 percent, as opposed to the 53.2 percent reported above. The 53.2 percent represents the overall 2012 ATUS sample, whereas the 54.0 percent reflects the response rate among the cases included in this analysis.

5. In multiple-adult households, it was possible that the CPS and CE Supplement respondent was not the same individual sampled for ATUS. In this case, social role and activity data were unknown for the ATUS sampled individual. These cases were excluded from the analyses. This will slightly limit the generalizability of the analysis, as the average household size among respondents will be smaller than the general population.

Table 1. CPS Question Wording of Social Activities and Social Roles by Route to Integration (ATUS)

Type of integration	Label	Question wording	Original responses	Analyzed responses
Other Engagement	Dinner w/family	How often did you eat dinner with any of the other members of your household? Basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Almost daily Few times/week Few times/month Monthly or less
	Friend/Family	During the past twelve months, how often did you see or hear from friends or family, whether in person or not? Basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Almost daily Few times/week Few times/month Monthly or less
	Parent	Parent (constructed from roster based on presence of children in household)	Parent Not a parent	Parent Not a parent
	Spouse	Spouse (constructed from roster)	Married Not married	Married Not married
	Sports group	Have you participated in any of these groups during the past 12 months, that is, since November 2010? A sports or recreation organization such as a soccer club or tennis club?	Yes No	Yes No

Continued

Table 1. *Continued*

Type of integration	Label	Question wording	Original responses	Analyzed responses
Other Engagement (cont'd)	Neighbor	How often did you talk with any of your neighbors? Basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Almost daily Few times/week Few times/month Monthly Less than monthly Never
	Employee	Last week, did you do any work for (either) pay (or profit)?	Yes No Retired Disabled Unable to work	Yes No
	Neighbor favors	How often did you and your neighbors do favors for each other? By favors we mean such things as watching each other's children, shopping, house sitting, lending garden or household items, and other small acts of kindness? Basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Almost daily Few times/week Few times/month Monthly Less than monthly Never

Continued

Table 1. ***Continued***

Type of integration	Label	Question wording	Original responses	Analyzed responses
Political/Civic Engagement	Talk politics	How often did you discuss politics with family or friends—basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Almost daily Few times/week Few times/month Monthly Less than monthly Never
	Vote	Do you always vote in local elections, do you sometimes vote, rarely vote, or do you never vote?	Always Sometimes Rarely Never	Always Sometimes Rarely Never
	Internet post	How often, if at all, have you used the Internet to express your opinions about POLITICAL or COMMUNITY issues within the past 12 months? Basically every day, a few times a week, a few times a month, once a month, less than once a month, or not at all?	Almost daily Few times/week Few times/month Monthly Less than monthly Never	Few times/week or more Few times/month Monthly Less than monthly Never
	Contact official	Please tell me whether or not you have done any of the following in the past 12 months, that is, between November 2010 and now. Contacted or visited a public official—at any level of government—to express your opinion?	Yes No	Yes No

Continued

Table 1. *Continued*

Type of integration	Label	Question wording	Original responses	Analyzed responses
Political/Civic Engagement (cont'd)	Boycott	Please tell me whether or not you have done any of the following in the past 12 months, that is, between November 2010 and now? Bought or boycotted a certain product or service because of the social or political values of the company that provides it?	Yes No	Yes No
	Other org.	Have you participated in any of these groups during the past 12 months, that is, since November 2010? Any other type of organization that I have not mentioned?	Yes No	Yes No
	Religious org.	Have you participated in any of these groups during the past 12 months, that is, since November 2010? A church, synagogue, mosque, or other religious institution or organization, NOT COUNTING your attendance at religious services?	Yes No	Yes No
	Civic org.	Have you participated in any of these groups during the past 12 months, that is, since November 2010? A service or civic organization such as American Legion or Lions Club?	Yes No	Yes No
	Community officer	In the past 12 months, that is, since November 2010, have you served on a committee or as an officer of any group or organization?	Yes No	Yes No
	Community group	Have you participated in any of these groups during the past 12 months, that is, since November 2010? A school group, neighborhood, or community association such as PTA or neighborhood watch group?	Yes No	Yes No

In addition to the 18 social activity and social role variables, the ATUS frame also included other variables that were used in the analyses, including the sample member's sex, race/ethnicity, household income, education, homeownership status, and age.

All analyses were conducted using the ATUS base weights and replicated base weights.

Survey of Health, Ageing, and Retirement in Europe (SHARE) Wave II

In order to test for generalizability across surveys, similar analyses were conducted using SHARE Wave II. SHARE is a longitudinal in-person survey that collects information on the health, economic, and social well-being of individuals aged 50 or older and their spouses. In order to be eligible for SHARE Wave I, sampled individuals must have been born before 1955, not have been institutionalized, speak the national language, and have a primary residence in the country from which they were sampled. All eligible individuals within a household were selected along with their spouses, regardless of the spouse's age (Börsch-Supan et al. 2013).

SHARE Wave I was conducted in 2004 in nine countries: Austria, Denmark, France, Germany, Greece, Italy, Spain, Switzerland, and the Netherlands.[6] The Wave I sampling frames varied by country based on availability and coverage. National or regional registers of individuals or households were used in some countries, while others used telephone frames. Due to variation in the sampling frames, sampling strategies also varied by country, with some using a simple random sample and others using a multistage clustered design. Finally, some countries did not have age information on the frame and had to conduct a screening interview to determine household eligibility prior to administering the main interview. Despite differences in sampling frames, each had nearly full coverage of the target population (Börsch-Supan and Jürges 2005).

A total of 31,036 sample lines were drawn in Wave I across the nine countries. Interviews were completed with 20,449 age-eligible individuals from 14,630 households, resulting in an average household-level AAPOR Response Rate 2 of 61.8 percent across countries.[7] The response rate varied by country, with

6. In some of the countries, data collection extended into 2005. SHARE Wave I was also conducted in Belgium, Israel, and Sweden, which we excluded from this paper because they operated on different time lags between waves and/or changed sampling designs between waves. Interviews were conducted in each country in the official language(s), with the exception of Switzerland, where Romansh is an official language but in which interviews were not conducted.

7. The response rate of 61.8 percent includes Sweden in addition to the nine countries we used. Not enough information was available in the Wave I dataset to recreate the response rate excluding Sweden. Country-level response rates were as follows—Denmark: 63.2 percent; Germany: 63.4 percent; Italy: 55.1 percent; the Netherlands: 61.3 percent; Spain: 53.3 percent; Austria: 58.1 percent; France: 73.6 percent; Greece: 61.4 percent; Switzerland: 37.6 percent.

most between 50 and 65 percent (Börsch-Supan and Jürges 2005). Attempts were made in 2006–2007 to reinterview the 20,449 Wave I respondents as part of Wave II. As with ATUS, if the most isolated individuals were nonrespondents in Wave I, then measures of social integration constructed from the Wave II frame would be positively skewed.

Analyses were limited to 19,299 of the original 20,449 sampled individuals (12,904 Wave II respondents and 6,395 nonrespondents). In order to be included, an individual had to have completed Wave I (i.e., proxy interviews were excluded) and have been born before 1955 (i.e., underage spouses were excluded). Individuals who died prior to Wave II were also excluded from analysis. The demographic makeup of the analytic subsample is nearly identical to the full Wave II sample in the nine countries.

A total of 12 social activities and roles questions were asked in Wave I and were available for all Wave II sample members, including six civic and political engagement measures (table 2). Estimates calculated from the Wave II sample could thus be compared to estimates calculated among Wave II respondents. Consistent with methods used to analyze the ATUS data, ordinal variables were sometimes collapsed into “never” and “at least once” categories for dichotomous analysis. In other instances, categories were collapsed when cell sizes were small. The categories used for analysis are shown in the last column of table 2.

A few of the social activity and role measures deserve note. First, many of the ordinal variables were created from a combination of questions. In some instances, interviewers asked whether or not an individual participated in an activity before asking about the frequency of participation. Individuals who said “no” to the base question were coded as “never.” In questions regarding frequency of contact with family members, questions were asked about each family member. In order to create a single variable for frequency of contact, an individual was assigned the most frequent category found across their family members. Second, some questions did not apply to some individuals. These individuals were dropped from the relevant analyses.

Additional socio-demographic variables were also available on the SHARE Wave II frame. These included country, household income, education, sex, age, homeownership status, employment status, and household size. These are slightly different from those available on ATUS. Race/ethnicity was not asked in SHARE Wave I and is not available for inclusion.

All analyses were conducted using individual-level (as opposed to household-level) Wave II base weights. These weights were calculated individually for each country based on the sample design, the probability of selection, and a nonresponse adjustment for Wave I nonresponse. A total of 72 replicated base-weights were also created using jackknife repeated replication (JRR).

Table 2. SHARE Wave I Question Wording of Social Activities and Social Roles by Route to Integration (SHARE)

Type of integration	Label	Question wording	Original responses	Analyzed responses
Political/Civic engagement	Volunteer	How often in the past four weeks did you do voluntary or charity work?	Almost daily Almost every week Less often Never	At least once Never
	Sick adult	How often in the past four weeks have you cared for a sick or disabled adult?	Almost daily Almost every week Less often Never	At least once Never
	Community group	How often in the past four weeks have you taken part in a political or community-related organization?	Almost daily Almost every week Less often Never	At least once Never
	Help others (derived from 3 questions)	In the past twelve months, have you personally given any kind of help listed on card 28 to a family member from outside the household, a friend, or a neighbor?	Yes No	Almost every month or more Less often Never
		Which [other] family member from outside the household, friend, or neighbor have you helped [most often] in the past twelve months?	Responses included: Friend (Ex-) colleague Neighbor Other acquaintance	
		In the past twelve months, how often altogether have you given such help to this person? Was it…	Almost daily Almost every week Almost every month Less often Never	

Continued

Table 2. ***Continued***

Type of integration	Label	Question wording	Original responses	Analyzed responses
Political/Civic engagement (cont'd)	Training	How often in the past four weeks have you attended an educational or training course? Almost daily, almost every week, less often?	Almost daily Almost every week Less often Never	At least once Never
	Religious org.	How often in the past four weeks have you taken part in a religious organization? Almost daily, almost every week, less often?	Almost daily Almost every week Less often Never	Almost every week or more Less often Never
Other engagement	Spouse/Partner	What is your marital status? Married and living together with spouse; registered partnership; and married, living separated from spouse; never married; divorced; widowed	Married, living together Registered partnership Married, living separate Never married Divorced Widowed	Married/ registered partnership Other

Continued

Table 2. ***Continued***

Type of integration	Label	Question wording	Original responses	Analyzed responses
Other engagement (cont'd)	Contact parent	During the past 12 months, how often did you have contact with your mother/father, either personally, by phone, or by mail? Daily, several times a week, about once a week, about every two weeks, about once a month, less than once a month, or never?	Daily Several times a week About once a week About every two weeks About once a month Less than once a month Never	Daily Several times a week About once a week Less often Never Missing (neither parent is alive)
	Contact child	During the past 12 months, how often did you or your husband/wife/partner have contact with [CHILD], either personally, by phone, or by mail? Daily, several times a week, about once a week, about every two weeks, about once a month, less than once a month, or never?	Daily Several times a week About once a week About every two weeks About once a month Less than once a month Never	Daily Several times a week Less often Never Missing (does not have children)
	Babysit	On average, how often did you look after the child(ren) of [CHILD] in the past twelve months? Almost daily, almost every week, less often?	Almost daily Almost every week Almost every month Less often Never	Almost daily Almost every week Almost every month Less often Never Missing (does not have grandchildren)

Continued

Table 2. ***Continued***

Type of integration	Label	Question wording	Original responses	Analyzed responses
Other engagement (cont'd)	Help HHM	Is there someone living in this household whom you have helped regularly during the past twelve months with personal care, such as washing, getting out of bed, or dressing?	Yes No	Yes No Missing (lives alone)
	Help family (derived from 3 questions)	In the past twelve months, have you personally given any kind of help listed on card 28 to a family member from outside the household, a friend, or a neighbor?	Yes No	Almost daily Almost every week Almost every month Less often Never
		Which [other] family member from outside the household, friend, or neighbor have you helped [most often] in the past twelve months?	The list is too long to include here. Any response was coded as family except those listed under "Help others"	
		In the past twelve months, how often altogether have you given such help to this person? Was it…	Almost daily Almost every week Almost every month Less often Never	

Methods

Hypothesis 1: Univariate Bias

Estimates of the 30 social activities and roles (18 in ATUS and 12 in SHARE) were constructed using the full sample and the respondent sample in order to test hypothesis 1 that univariate estimates of social activities and roles should be upwardly biased. The categorical variables were evaluated using both the coding scheme outlined in the last columns of tables 1 and 2 and as the dichotomies "never" versus "at least once."

A simple *t*-test or chi-squared test was not appropriate to test the differences between the full sample and the respondents because the two samples were not independent. In order to account for the covariance between them, replication was used to adjust the standard deviations. For dichotomous variables, both the absolute and relative differences of the proportion between the respondents and the full sample were calculated for each of the *I* base-weighted replicates ($I = 160$ and 72 replicates for ATUS and SHARE, respectively). The standard deviation of the absolute difference of proportions across all replicates may be written as follows:

$$d_i = p_{ri} - p_{fi} \tag{1}$$

$$s_d = \sqrt{\frac{1}{I}\left(\sum_{i=1}^{I}(d_i - (p_r - p_f))^2\right)} \tag{2}$$

where

d_i = the absolute difference of proportions between the estimate among respondents and the estimate among the full sample for replicate i

p_r = the proportion of respondents that participated in the social activity/holds the social role

p_f = the proportion of the full sample that participates in the social activity/holds the social role

s_d = the standard deviation of the absolute difference of proportions of the estimate across all replicates

The *t*-statistic was calculated using the standard deviation of the difference across all replicates:

$$t_{I-1} = \frac{p_r - p_f}{s_d} \tag{3}$$

Similarly, the equations for adjusting the standard deviations of the relative difference are

$$d_i^* = \frac{p_{ri} - p_{fi}}{p_{fi}} \tag{4}$$

$$s_d^* = \sqrt{\frac{1}{I}\left(\sum_{i=1}^{I}\left(d_i^* - \frac{p_r - p_f}{p_f}\right)^2\right)}$$

$$t^*_{I-1} = \frac{p_r - p_f}{s^*_d}$$

where the "*" indicates the statistic was calculated from the relative difference.

Replication was also used to test the distributions of the categorical variables. Similar to a traditional chi-squared test, the difference of the proportion of cases that fell into each response category was calculated between the respondents (i.e., observed) and the full sample (i.e., expected) and divided by the adjusted variance. This set up the following equation for each variable:

$$\chi^2_{m-1} = (\mathbf{p_r} - \mathbf{p_f})^T (\mathbf{V}(\mathbf{p_r} - \mathbf{p_f}))^{-1} (\mathbf{p_r} - \mathbf{p_f}) \tag{5}$$

where $(\mathbf{p_r} - \mathbf{p_f})$ is an M_j by 1 vector where M_j is the number of response categories for variable j. The variance is an M_j by M_j matrix defined as

$$\mathbf{V}(\mathbf{p_r} - \mathbf{p_f}) = \frac{1}{I}\sum_{i=1}^{I}((\mathbf{p_{ri}} - \mathbf{p_{fi}})(\mathbf{p_{ri}} - \mathbf{p_{fi}})^{\mathbf{T}}) \tag{6}$$

Given the number of tests performed (43 across both samples, all variables, and all tests) and a 95 percent confidence interval, two were expected to be significant by chance. To minimize the potential of making a Type I error, the false discovery rate (FDR) was used to adjust the conclusions drawn from the p-values (Benjamini and Hochberg 2000). This approach is similar to a Bonferroni adjustment in that it accounts for the increased chance of committing a Type I error as the number of tests increases. However, a Bonferroni adjustment increases Type II errors and may cause researchers to draw different conclusions solely based on the number of tests run. The FDR does not have either of these weaknesses. While the FDR assumes independent samples across tests, simulation studies have suggested that the FDR is just as effective when this assumption is violated, as was the case here (Benjamini and Yekutieli 2001).

Hypothesis 2: Comparative Bias

In order to examine hypothesis 2—estimates of civic and political activities and roles should be more biased than estimates of other social activities and roles—the dichotomous form of the variables (18 in ATUS and 12 in SHARE) was sorted by the magnitude of both the absolute and relative bias and the results reviewed.

In order to statistically test the hypothesis, the difference of the bias among each combination of two variables was compared. Replication was used to control for the covariance between the full sample and respondents. The equations used to perform the t-tests were similar to those used to assess univariate bias, with one exception. In the univariate analysis, the t-tests were conducted on the difference between the full sample and the respondents, d_i. In this analysis, the t-tests were used to test the difference of the absolute and relative biases between two variables. The difference of the absolute bias may be calculated as

$$h_i = d_{ji} - d_{j'i} \tag{7}$$

$$s_h = \sqrt{\frac{1}{I}(\sum_{i=1}^{I}(h_i - (d_j - d_{j'}))^2)}$$

$$t_{I-1} = \frac{d_j - d_{j'}}{s_h}$$

and of the relative bias as

$$h_i^* = d_{ji}^* - d_{j'i}^* \tag{8}$$

$$s_h^* = \sqrt{\frac{1}{I}(\sum_{i=1}^{I}(h_i^* - (d_j^* - d_{j'}^*))^2)}$$

$$t_{I-1}^* = \frac{d_j^* - d_{j'}^*}{s_h^*}$$

where d^*_{ji} ($d^*_{j'i}$) is the difference in an estimate between the respondent sample and the full sample for the variable j (j') in replicate i. If the test was significant and if $d_j^* - d_{j'}^*$ was greater than zero, then we concluded that variable j was significantly more biased than variable j'. A total of 153 ($_{18}C_2$) comparisons were made in ATUS and 66 ($_{12}C_2$) in SHARE. The FDR was calculated and used to control for Type I error.

Hypothesis 3: Multivariate Bias

Sociologists, economists, and others frequently use logit models to predict social activities and roles as a function of a variety of demographic variables (e.g., Levin-Waldman 2013; McCabe 2013; Wemlinger and Kropf 2013). While other, more elaborate, models of social behavior are also found in the literature, we were limited to those that could be replicated using the ATUS and SHARE frame data. Therefore, we used a logit similar to those cited above. For ATUS, each social activity or role was regressed on homeownership, race/ethnicity, education, marital status, sex, age, employment status, presence of children in the household, and household income. Race/ethnicity and presence of children were not asked of SHARE respondents, so country of residence and household size were used instead.

The dependent variable was the dichotomous version of each of the social activities and roles, with the exception of activities and roles that were included as independent variables in the model (e.g., spouse). Models were built using

15 different dependent variables from ATUS and 11 from SHARE. The model was run twice for each dependent variable, once using data from the full sample and once limiting the analysis to respondents.

The models were rerun across each replicate, resulting in a total of 320 independent model runs for each ATUS dependent variable (160 replicates * 2 models per dependent variable) and 144 for each SHARE dependent variable (72 replicates * 2 models per dependent variable). Since coefficients in a given model are correlated with each other, comparisons of changes in the coefficients across two models is typically not possible. By computing replicate versions of the vector of beta coefficients, we implicitly reflect both the complexities of the estimation process (e.g., stratification, clustering, weighting, and form of the model) and the fact that the coefficients are correlated with each other (Krewski and Rao 1981; Binder 1983), allowing a direct comparison of coefficients between the full sample and respondent models.

The model constructed from the full sample was compared to the model constructed using the respondents on two metrics. First, *t*-tests were used to determine whether corresponding beta coefficients were significantly different from one another. The difference of the corresponding beta coefficients was calculated for each replicate as

$$b_{xi} = \beta_{rxi} - \beta_{fxi} \tag{9}$$

where *x* represents each of the independent variables in the model. The standard deviation of the difference and the *t*-statistic were calculated similarly to the other analyses using replication:

$$s_{bx} = \sqrt{\frac{1}{I}\left(\sum_{i=1}^{I}(b_{xi} - (\beta_{rx} - \beta_{fx}))^2\right)} \tag{10}$$

$$t_{I-1} = \frac{\beta_{rx} - \beta_{fx}}{s_{bx}}$$

A total of 255 comparisons (17 coefficients * 15 models) were made in ATUS and 209 (19 coefficients * 11 models) in SHARE. FDR was used to account for these multiple tests.

Second, a qualitative analysis was undertaken to determine whether corresponding models were likely to result in similar conclusions. The significance levels of corresponding beta coefficients were compared. Coefficients that had the same or similar significance levels would likely be interpreted similarly. For example, if age was significant at the 0.001 level in both the full sample model and the respondent model, then one would likely draw a similar conclusion about the effect of age on the outcome variable, regardless of the difference in the magnitude.

Results

Hypothesis 1: Univariate Bias

Tables 3 and 4 display the level of nonresponse bias identified in the social activity and role variables in ATUS and SHARE, respectively. Bias was observed in the expected direction for 24 of the 30 dichotomous variables, lending support for hypothesis 1. These findings held even after applying the FDR. In many cases, the differences were large, with up to a 23.6 percent relative and 4.2 percentage point absolute change in the estimate, but some of the differences were small and negligible. For example, 98.0 percent of the ATUS sample communicated with friends or family. Limiting the analysis to respondents increased the estimate to 98.4 percent. Although both the relative and absolute differences in the estimate were significant ($p = 0.0002$), the magnitude of the bias was so small as to be of little concern to most researchers.

Six variables, three in each survey, did not behave as expected. The proportion of employees (ATUS) and the proportion of individuals who contact their children (SHARE) were unbiased. Four other variables—the proportion of parents (ATUS), individuals who have dinner with family (ATUS), individuals who help household members (SHARE), and individuals who are in communication with their parents (SHARE)—were all lower among respondents than for the full sample. The bias on the proportion of parents (31.6 percent vs. 29.0 percent among the full sample and respondents, respectively) is inconsistent with previous research that found parents are more likely to respond (for a summary of the literature, see Groves and Couper 1998). Thus, it seemed possible that the ATUS finding was an artifact of the selection criteria (individuals had to be both the CE Supplement respondent and sampled for ATUS) that resulted in the inclusion of a disproportionate number of single-adult households and, consequently, single-parent households. Single-parent households may be busier or otherwise different from dual-parent households, reducing their propensity to respond. However, the proportion of parents among respondents remained significantly lower than the full sample, even after controlling for the number of adults in the household (results not shown).

The categorical analysis for both surveys mimicked the findings of the dichotomous analysis. All 13 of the categorical distributions were significantly different between the full sample and respondents (tables 5 and 6 for ATUS and SHARE, respectively). Twelve of the differences were in the expected direction, with respondents reporting more frequent participation than the full sample. Having dinner with family (ATUS) was the only exception. Respondents were more apt to be at both ends of the distribution almost daily and never. While the chi-squared tests were significant in all comparisons across both surveys, the differences observed in SHARE were, in some cases, smaller than those in ATUS. As with the dichotomous analysis, one should factor in the magnitude of the differences along with the significance levels. For example, SHARE respondents reported more frequent religious participation than the full sample.

Table 3. Differences in Social Activity and Role Estimates by Sample Type (ATUS)

	N		Sample type		Relative difference		Absolute difference	
	Full sample	Respondents	Full sample	Respondents	Value	*p*-value	Value	*p*-value
Parent	5,150	2,779	31.6%	29.0%	–8.04%	<0.0001	–2.54	<0.0001
Dinner w/family	5,009	2,732	76.3%	74.8%	–2.01%	<0.0001	–1.54	<0.0001
Employee	5,148	2,778	56.5%	56.3%	–0.46%	0.249	–0.26	0.248
Family/Friend	4,925	2,707	98.0%	98.4%	0.40%	0.0002	0.39	0.0002
Neighbor	4,925	2,705	89.2%	91.1%	2.12%	<0.0001	1.90	<0.0001
Neighbor favors	4,895	2,689	67.2%	70.6%	5.07%	<0.0001	3.40	<0.0001
Community group†	4,989	2,723	17.4%	18.2%	5.07%	0.001	0.88	0.001
Vote†	5,035	2,745	72.8%	76.7%	5.40%	<0.0001	3.93	<0.0001
Talk politics†	4,921	2,697	76.0%	80.3%	5.56%	<0.0001	4.23	<0.0001
Spouse	5,150	2,779	49.5%	52.9%	6.83%	<0.0001	3.38	<0.0001
Internet post†	4,975	2,714	28.0%	30.2%	7.95%	<0.0001	2.22	<0.0001
Religious org.	4,978	2,719	22.6%	25.9%	15.03%	<0.0001	3.39	<0.0001
Boycott†	4,996	2,728	13.0%	15.2%	17.60%	<0.0001	2.28	<0.0001
Sports group	4,989	2,723	11.2%	13.3%	18.99%	<0.0001	2.13	<0.0001
Contact official†	5,007	2,734	14.6%	17.4%	19.34%	<0.0001	2.82	<0.0001
Community officer†	4,983	2,723	13.0%	15.6%	19.65%	<0.0001	2.56	<0.0001
Other org.†	4,983	2,721	6.4%	7.7%	21.28%	<0.0001	1.35	<0.0001
Civic org.†	4,987	2,722	8.8%	10.8%	23.55%	<0.0001	2.07	<0.0001

†Identifies civic and political activities and roles.

Table 4. Differences in Social Activity and Role Estimates by Sample Type (SHARE)

	N		Sample type		Relative difference		Absolute difference	
	Full sample	Respondents	Full sample	Respondents	Value	*p*-value	Value	*p*-value
Help HHM	15,115	10,145	7.0%	6.9%	–1.57%	<0.0001	–0.11	<0.0001
Contact parent	4,993	3,386	96.9%	96.5%	–0.48%	<0.0001	–0.47	<0.0001
Contact children	16,851	11,454	99.3%	99.3%	0.00%	0.099	0.00	0.100
Spouse	19,229	12,890	66.6%	68.5%	2.79%	<0.0001	1.86	<0.0001
Babysit	11,116	7,546	47.1%	49.9%	5.99%	<0.0001	2.82	<0.0001
Religious org.†	19,091	12,864	9.5%	10.1%	6.37%	<0.0001	0.61	<0.0001
Help family	19,097	12,864	20.4%	21.7%	6.60%	<0.0001	1.35	<0.0001
Sick adult†	19,091	12,864	5.2%	5.5%	7.06%	<0.0001	0.36	<0.0001
Help others†	19,097	12,864	10.9%	11.7%	7.98%	<0.0001	0.87	<0.0001
Training	19,090	12,863	4.3%	4.7%	9.55%	<0.0001	0.41	<0.0001
Volunteer†	19,090	12,863	10.1%	11.6%	14.93%	<0.0001	1.51	<0.0001
Community group†	19,089	12,863	3.2%	3.8%	16.80%	<0.0001	0.54	<0.0001

†Identifies civic and political activities and roles.

Table 5. Differences in Social Activity and Role Distributions by Sample Type (ATUS)

		Sample type		Absolute	
		Full sample	Resp.	difference	X^2 (*p*-value)
Dinner w/ family	Monthly or less	25.3%	26.4%	1.11	
	Few times/month	3.6%	3.0%	–0.64	
	Few times/week	13.6%	12.6%	–0.92	
	Almost daily	57.6%	58.0%	0.45	
	TOTAL N	5,009	2,732		42.1 (<0.0001)
Friend/ Family	Monthly or less	7.9%	7.3%	–0.61	
	Few times/month	12.3%	11.8%	–0.54	
	Few times/week	36.0%	35.9%	–0.05	
	Almost daily	43.7%	44.9%	1.19	
	TOTAL N	4,925	2,707		22.0 (<0.0001)
Neighbor	Never	10.8%	8.9%	–1.90	
	Less than monthly	9.3%	8.6%	–0.71	
	Monthly	9.0%	9.1%	0.12	
	Few times/month	24.2%	25.0%	0.77	
	Few times/week	33.3%	35.0%	1.71	
	Almost daily	13.4%	13.4%	0.01	
	TOTAL N	4,925	2,705		95.7 (<0.0001)
Neighbor favors	Never	32.8%	29.4%	–3.40	
	Less than monthly	20.8%	22.2%	1.42	
	Monthly	13.3%	14.3%	1.00	
	Few times/month	19.3%	20.6%	1.27	
	Few times/week	10.5%	10.4%	–0.06	
	Almost daily	3.3%	3.1%	–0.23	
	TOTAL N	4,895	2,689		134.4 (<0.0001)

Continued

Table 5. *Continued*

		Sample type		Absolute	
		Full sample	Resp.	difference	X^2 (*p*-value)
Vote†	Never	27.2%	23.3%	–3.93	
	Rarely	9.6%	9.3%	–0.30	
	Sometimes	27.2%	27.5%	0.26	
	Always	36.0%	39.9%	3.97	
	TOTAL N	5,035	2,745		204.0 (<0.0001)
Talk politics†	Never	24.0%	19.7%	–4.23	
	Less than monthly	15.4%	15.5%	0.11	
	Monthly	10.8%	11.2%	0.38	
	Few times/month	20.4%	21.5%	1.08	
	Few times/week	19.8%	21.6%	1.84	
	Almost daily	9.6%	10.4%	0.81	
	TOTAL N	4,921	2,697		235.0 (<0.0001)
Internet post†	Never	72.0%	69.8%	–2.22	
	Less than monthly	11.3%	12.6%	1.36	
	Monthly	4.5%	5.0%	0.52	
	Few times/month	5.4%	6.1%	0.68	
	Few times/week or more	6.9%	6.5%	–0.33	
	TOTAL N	4,975	2,714		55.9 (<0.0001)

†Identifies civic and political activities and roles.

Table 6. Differences in Social Activity and Role Distributions by Sample Type (SHARE)

		Sample type		Absolute	
		Full sample	Resp.	difference	X^2 (*p*-value)
Contact parent	Never	3.5%	3.1%	–0.47	
	Every 2 weeks or less	18.0%	18.1%	0.18	
	About once per week	21.6%	21.8%	0.16	
	Several times per week	27.3%	28.0%	0.72	
	Daily	29.6%	29.0%	–0.60	
	TOTAL N	4,993	3,386		782.5 (<0.0001)
Contact children	Never	0.7%	0.7%	–0.02	
	Weekly or less	15.4%	13.9%	–1.50	
	Several times per week	22.0%	21.6%	–0.46	
	Daily	61.9%	63.8%	1.98	
	TOTAL N	16,851	11,454		1,066.3 (<0.0001)
Religious org.[†]	Never	90.5%	89.9%	–0.61	
	Less than weekly	3.5%	3.8%	0.30	
	Almost every week or more	6.0%	6.3%	0.31	
	TOTAL N	19,091	12,864		575.4 (<0.0001)
Help family	Never	79.6%	78.3%	–1.34	
	Less than monthly	5.0%	5.4%	0.33	
	Almost every month	4.1%	4.3%	0.15	
	Almost every week	6.4%	6.7%	0.26	
	Almost daily	4.8%	5.4%	0.61	
	TOTAL N	19,097	12,864		1,605.3 (<0.0001)

Continued

Table 6. *Continued*

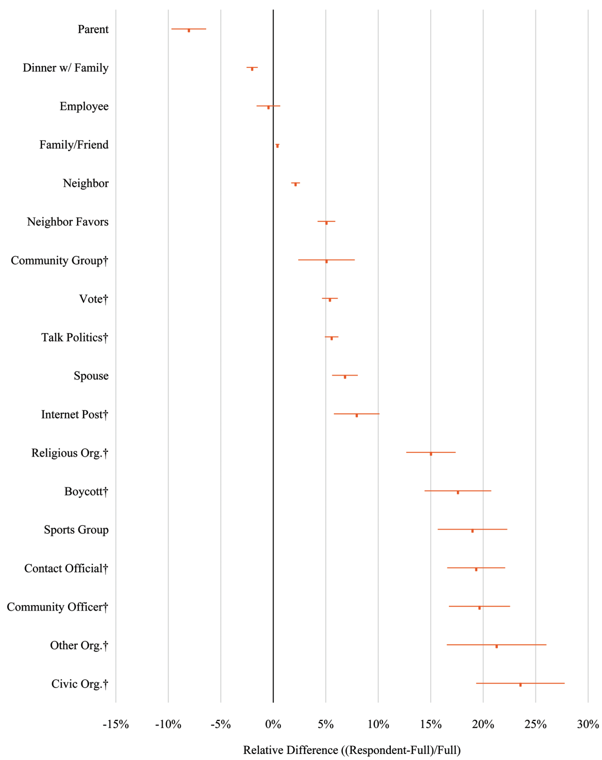
		Sample type		Absolute	
		Full sample	Resp.	difference	X^2 (*p*-value)
Help others[†]	Never	89.1%	88.3%	–0.87	
	Less than monthly	4.6%	5.0%	0.40	
	Almost every month or more	6.3%	6.8%	0.47	
	TOTAL N	19,097	12,864		827.1 (<0.0001)
Babysit	Never	53.0%	50.2%	–2.85	
	Less than monthly	12.2%	12.9%	0.78	
	Almost every month	8.5%	8.4%	–0.06	
	Almost every week	14.6%	15.6%	1.03	
	Almost daily	11.7%	12.8%	1.10	
	TOTAL N	11,116	7,546		2,391.7 (<0.0001)

[†]Identifies civic and political activities and roles.

However, the differences were no larger than 0.61 percentage points in any given response category, suggesting inconsequential bias.

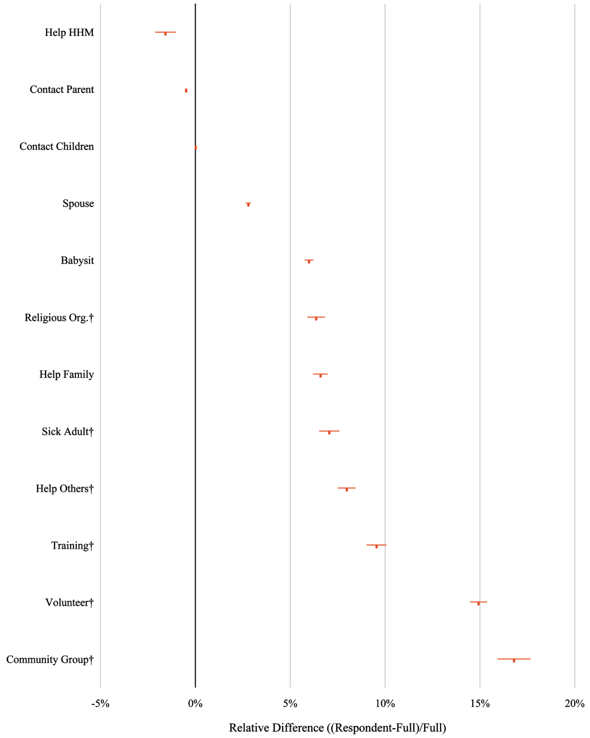
Hypothesis 2: Comparative Bias

In order to test hypothesis 2, variables were ordered by both their relative and absolute difference of proportions. For both surveys, variables generally clustered into three levels of relative bias: less than 5 percent, 5 to 10 percent, and more than 10 percent (see figures 1 and 2 for ATUS and SHARE, respectively). In both surveys, all the civic and politically oriented variables fell into the two higher categories. However, the confidence intervals of the estimates of the civic and political activities and roles frequently overlapped with those of the other variables. For example, the confidence interval of ATUS's community group measure was not significantly different from three of the eight variables that measured something other than a civic or political activity or role (neighbor, neighbor favors, and spouse). Significance tests performed on



† Identifies civic and political activities and roles.

Figure 1. Relative Difference in Univariate Estimates ([Respondents–Full Sample]/Full Sample) (ATUS).

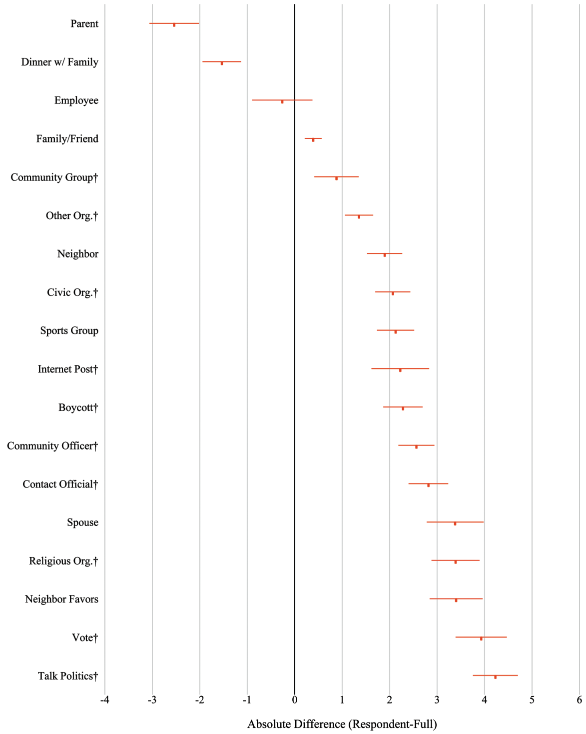


† Identifies civic and political activities and roles.

Figure 2. Relative Difference in Univariate Estimates ([Respondents–Full Sample]/Full Sample) (SHARE).

the differences between variables (results not shown) were consistent with the pictorial representation. These results show that while civic and politically oriented variables trended toward higher levels of relative bias, the differences were not statistically significant.

Reordering the data by absolute difference told a somewhat different story. The civic and political variables were interspersed throughout the range of values, suggesting no relationship between civic engagement and level of bias (see figures 3 and 4 for ATUS and SHARE, respectively). The discrepancies between the findings of the relative and absolute analysis were a result of the differences in the point estimates across item types. Civic and political activities and roles were generally less common than other types of activities and roles. As a result, similar absolute differences resulted in larger relative differences among the civic and political measures. Overall, there was some evidence for hypothesis 2, but findings were not significant nor consistent between analysis methods.



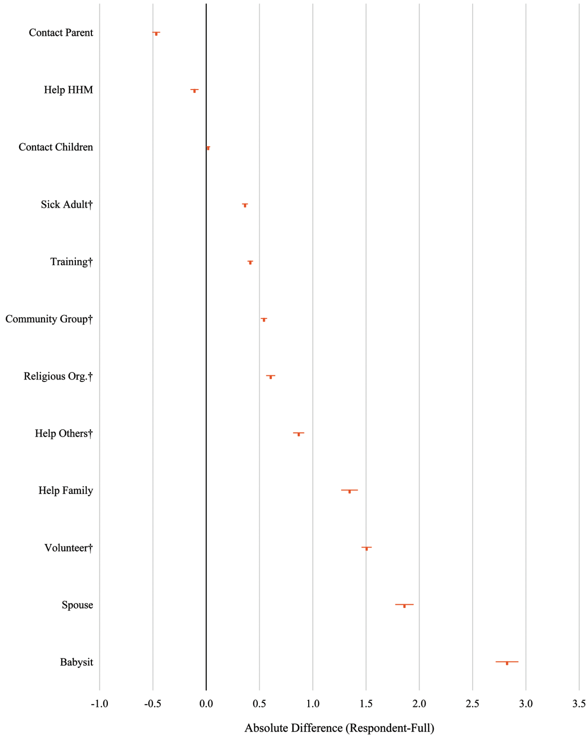
† Identifies civic and political activities and roles.

Figure 3. Absolute Difference of Univariate Estimates (Respondents–Full Sample) (ATUS).

Hypothesis 3: Multivariate Bias

To test hypothesis 3—coefficients of independent variables in multivariate models that predict social activities and roles should be unbiased—logit models were estimated in both ATUS and SHARE.[8] The models themselves generally had a fair model fit, with areas under the receiver operating characteristic (ROC) curve hovering around 0.70. A total of 464 comparisons were made between coefficients of models that used the full sample and those that used just the respondents. Given the large number of comparisons, a total of 23 comparisons would have been significant at the 5 percent level by chance. However, 256 (55.2 percent, after applying the FDR) of the beta coefficients in the respondent models were significantly different from their full-sample counterparts. The proportion of significant differences was much smaller in the ATUS models (29.0 percent) than in SHARE's (87.1 percent). As with the

8. Given the large number of models and the inconsequentiality of any one coefficient, the full models are presented in the online supplement (online table A and online table B).



† Identifies civic and political activities and roles.

Figure 4. Absolute Difference of Univariate Estimates (Respondents–Full Sample) (SHARE).

previous analyses, much of the difference in the number of significant findings was the result of tighter confidence intervals for the SHARE coefficients. Among significant differences, the magnitude of the difference was frequently smaller in SHARE than in ATUS. Even after considering the small magnitudes of some of the differences, there was still ample evidence to suggest that nonresponse bias was not limited to univariate analyses.

The beta coefficients were examined in a variety of ways in an attempt to identify a pattern of bias. First, the direction of the bias was examined. However, the direction of the bias was inconsistent across variables and surveys. This was not unexpected since the relationship between the dependent and independent variable could have been positive or negative, and the effect of nonresponse could have accentuated or mediated the relationship.

Second, the number of significantly different betas was examined by dependent variable (see tables 7 and 8 for ATUS and SHARE, respectively). Models predicting some social activities and roles were more likely to result in

Table 7. Number of Significant Differences Found in the Multivariate Models in Online Table A by Significance Level and Dependent Variable (ATUS)

	p-value			
	<0.001	<0.01	<0.05	n.s.
Civic org.†	0	2	0	15
Dinner w/family	1	1	0	15
Committee officer†	1	1	0	15
Vote†	1	2	0	14
Religious org.†	2	1	0	14
Friend/Family	1	3	1	12
Talk politics†	0	2	3	12
Contact official†	2	2	1	12
Neighbor favors	2	3	0	12
Other org.†	3	2	0	12
Sports group	4	0	1	12
Community group†	2	4	1	10
Boycott†	0	6	1	10
Internet post†	2	7	0	8
Neighbor	5	3	1	8

†Identifies civic and political activities and roles.

Table 8. Number of Significant Differences Found in the Multivariate Models in Online Table B by Significance Level and Dependent Variable (SHARE)

	p-value			
	<0.001	<0.01	<0.05	n.s.
Sick adult†	11	1	1	6
Religious org.†	11	0	3	5
Help family	11	0	3	5
Babysit	14	1	2	2
Help HHM	15	1	1	2
Community group†	16	1	0	2
Contact children	16	2	0	1
Contact parent	15	3	0	1
Help others†	16	1	1	1
Training†	16	2	0	1
Volunteer†	17	1	0	1

†Identifies civic and political activities and roles.

significantly different coefficients between samples. For example, the ATUS model predicting civic organization membership had many more unbiased coefficients than the model predicting communication with a neighbor (18 vs. eight unbiased coefficients). No clear pattern emerged on the type of dependent variable that suffered from biased coefficients.

Finally, bias was examined by independent variable (see tables 9 and 10 for ATUS and SHARE, respectively). For ATUS, marital status was found to be biased more often than other variables, with eight significantly different coefficients. Age and having children in the household were least likely to have a biased coefficient, with only one significantly different coefficient across the 15 models. The number of significant differences was more consistent across the SHARE independent variables. All variables were observed to have between eight and 11 significant differences. As with the previous analysis, no clear patterns were identified.

It was possible for a coefficient to have a significantly different magnitude between models, but have similar significance levels. For example, the effect

Table 9. Number of Significant Differences Found in the Multivariate Models in Online Table A by Significance Level and Independent Variable (ATUS)

	p-value			
	<0.001	<0.01	<0.05	n.s.
Intercept	1	2	1	11
Homeowner	2	3	0	10
Race/Ethnicity (ref = NH White)				
NH Black	2	1	0	12
Hispanic	1	5	0	9
NH Other	2	4	1	8
Education (ref = LT HS)				
High school	3	1	1	10
Some college	2	2	0	11
College degree or more	2	3	0	10
Married	3	3	2	7
Female	2	5	0	8
Age	1	0	0	14
Employed	1	3	2	9
Children in household	0	0	1	14
Income (ref = LT $20k)				
$20,000–$39,999	1	2	0	12
$40,000–$59,999	0	2	1	12
$60,000–$99,999	1	2	0	12
$100,000 or more	2	1	0	12

Table 10. Number of Significant Differences Found in the Multivariate Models in Online Table B by Significance Level and Independent Variable (SHARE)

	p-value			
	<0.001	<0.01	<0.05	n.s.
Intercept	9	0	0	2
Homeowner	9	0	1	1
Country (ref = Austria)				
Germany	10	1	0	0
The Netherlands	8	0	2	1
Spain	8	0	0	3
Italy	9	1	0	1
France	7	2	1	1
Denmark	7	1	0	3
Greece	9	0	0	2
Switzerland	9	0	0	2
Education (ref = Primary school or less)				
Some secondary school	7	1	1	2
Secondary school	9	0	0	2
First stage tertiary or higher	10	1	0	0
Married	9	0	1	1
Female	9	0	1	1
Age	9	1	0	1
Employed	7	2	1	1
HH Size	5	4	2	0
Income (per 1,000€)	8	0	1	2

of being Hispanic on voting was significantly larger in the ATUS respondent model than in the full-sample model (–0.71 vs. –0.59, respectively), but both coefficients were significantly predictive of voting at the 0.001 level. By contrast, although the magnitude of the effect of being non-Hispanic Black on membership in a community organization was unchanged across the full sample and respondent samples, the coefficient was significant at the 0.001 level in the full model but was not significant in the respondent model. In the first example, a researcher would likely draw the same conclusion about the effect of being Hispanic on voting even though the magnitude of the effect was biased. In the second example, a researcher would likely draw a different conclusion about the effect of being non-Hispanic Black even though the magnitude of the effect was unbiased. Across the two surveys, the significance level of the coefficients was observed to be consistent between the full sample and respondent models 72.8 percent (n = 338) of the time. Among the 126 coefficients that changed significance level between models, the vast majority of differences (69.0 percent) were marginal. That is, they changed from one level of significance (e.g., 0.01) to a neighboring level (e.g., 0.05 or 0.001). Thus, although overall, the coefficients of the multivariate models were biased, resulting in no support for hypothesis 3, the bias was generally unlikely to change the interpretation of the models since the significance level of the coefficients was often unchanged between models.

Discussion

Of 507 tests performed to identify nonresponse bias, 297 (58.6 percent) yielded significant results, far too many to be attributed to chance. While the magnitude of the bias varied significantly across items and surveys, the theme was consistent: nonresponse bias existed in both surveys, in univariate estimates (H1) and in multivariate models (H3). Of the 30 univariate estimates for which bias was assessed, 24 were in the expected direction: the respondent estimate was higher than the full sample estimate. Bias was observed in all multivariate models, although the number of biased coefficients differed by survey, dependent variable, and independent variable. Unlike the univariate bias, the bias observed in the multivariate models was often small and rarely changed the interpretation of the model. Given the limited fit and associated noise of these models, the bias may be smaller than would be observed under a better-fitting model.

The multivariate results extend the findings of Abraham, Helms, and Presser (2009). They found that volunteering was more common among respondents than nonrespondents, across many different subgroups, suggesting that similar conclusions about the link between volunteering and subgroup membership would be drawn from the respondent and full-sample data. But Abraham et al. did not compare the associations between volunteering and subgroup membership for the respondents to those for the full sample. When we did that in our surveys, we found differences large enough to bias some coefficients. However, consistent with Abraham et al., the bias was often small and unlikely to alter interpretations.

What differentiates the 297 biased statistics from the remaining 210? Univariate estimates of civic and political indicators trended toward higher levels of nonresponse bias, but the differences were not significant and the trend did not hold for the multivariate analyses. We were unable to discover a pattern distinguishing the significant from the nonsignificant results.

While bias was identified in both surveys, its magnitude was generally larger for ATUS comparisons, though more significant differences were identified in SHARE. The number of significant differences found in SHARE is in part due to the small confidence intervals associated with the large sample sizes. The difference in the magnitude of the bias across surveys may be the result of differences in response rates and frame quality. SHARE Wave II achieved a higher response rate than ATUS. Since nonresponse bias is a function of both response rate and the difference between respondents and nonrespondents, the difference would have had to be larger in the SHARE sample than the ATUS sample in order to observe the same bias. Moreover, the ATUS sample was drawn from CPS-responding households while the SHARE Wave II sample was drawn from the Wave I respondents. At the time of ATUS recruitment, the individual had been asked to participate in eight waves of the CPS over the past 14 months whereas SHARE Wave II individuals had only one prior request. The Bureau of Labor Statistics (2015) has noted that survey fatigue may be the primary reason for nonresponse to ATUS. If respondent fatigue is correlated with social integration, that could explain some of the differences between surveys in the magnitude of the effect of the bias. Additionally, neither the CPS nor SHARE Wave I had 100 percent response rates. If the most isolated respondents did not participate in the CPS or SHARE Wave I, then the sample estimates would themselves be biased, and less difference would be observed between the full-sample estimate and the respondent estimate. Since SHARE Wave I had a much lower response rate than the CPS, the magnitude of the difference would be smaller for SHARE than ATUS. While these explanations of the differences between ATUS and SHARE are plausible, they are untestable since information on the CPS and SHARE Wave I nonrespondents is unavailable.

Our findings underscore the need for further research to better identify the nature of the link between social integration and nonresponse. But they also highlight the need to develop both data-collection methods that obtain response from a more representative sample of individuals (by targeting efforts disproportionately among the less integrated) and data-adjustment methods that better correct for nonresponse bias (by incorporating factors other than standard background factors in weighting procedures).

Supplementary Data

Supplementary data are freely available online at http://poq.oxfordjournals.org/

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